CONTRACTIBILITY CAN FAIL AT THE TOP OF THE RANK-SPREAD FILTRATION: A TEN-ELEMENT GRADED POSET

Abstract. Let P be a finite graded poset of rank n and let $\Delta^{(k)}(P)$ be the subcomplex of the order complex $\Delta(P)$ spanned by the chains of rank spread at most k, so that $\Delta^{(0)}(P)$ is the vertex set and $\Delta^{(n)}(P) = \Delta(P)$. Kitajima asks whether $\Delta^{(k)}(P)$ is contractible whenever $\Delta^{(k-1)}(P)$ is. We exhibit a counterexample to that implication as stated. Let P be the poset on the ten elements $a_0, a_1, b_0, b_1, b_2, c_0, c_1, c_2, d_0, d_1$, of ranks $0, 0, 1, 1, 1, 2, 2, 2, 3, 3$, whose covers are exactly the fifteen relations displayed in Section 2. Then P is graded of rank 3, the complex $\Delta^{(2)}(P)$ collapses to a point in 27 elementary steps, and $\Delta^{(3)}(P) = \Delta(P)$ is homotopy equivalent to S^2; in particular $\widetilde{H}_2(\Delta^{(3)}(P); \mathbb{Z}) = \mathbb{Z}$ and $\Delta^{(3)}(P)$ is not contractible. The witness has $k = n$, so what fails to be contractible is the full order complex; whether the implication can fail at a proper stage $k < n$ we leave open.

1. The question

Let P be a finite poset. Its *order complex* $\Delta(P)$ is the simplicial complex whose faces are the nonempty chains of P. Following Kitajima [3], P is *graded of rank* n if every maximal chain of P has length n — that is, P is *pure*; no bound is imposed, so P may have several minimal and several maximal elements. A graded poset carries a rank function $\mathrm{rk}\colon P \to \{0, \dots, n\}$, the *rank spread* of a chain $C \subseteq P$ is $\max_{x \in C} \mathrm{rk}(x) - \min_{x \in C} \mathrm{rk}(x)$, and for $0 \le k \le n$

$$\Delta^{(k)}(P) \;:=\; \{\, C \in \Delta(P) \;:\; \text{the rank spread of } C \text{ is at most } k \,\}$$

is a subcomplex of $\Delta(P)$. Thus $\Delta^{(0)}(P)$ is the vertex set, $\Delta^{(1)}(P)$ is the Hasse diagram viewed as a 1-complex,

$$\Delta^{(0)}(P) \subseteq \cdots \subseteq \Delta^{(n)}(P) = \Delta(P),$$

and the filtration is the poset case of the filtered nerve of Di et al. [1], whose relative homology $H_*(\Delta^{(k)}(P), \Delta^{(k-1)}(P))$ is the magnitude homology of P [1, Lemma 1.7]; the order-complex antecedent for metric spaces is due to Kaneta and Yoshinaga [2]. In that context Kitajima records, verbatim:

> "It is an open problem whether $\Delta^{(k)}(P)$ is contractible whenever $\Delta^{(k-1)}(P)$ is contractible."

2020 *Mathematics Subject Classification.* 06A07, 55U10, 05E45.

Key words and phrases. graded poset, order complex, rank-spread filtration, contractibility, collapsibility, magnitude homology, counterexample.

The sentence is in Remark 2.13 on page 8 of [3] (version v2 of 24 June 2026); the ambient definition of "graded of rank n" as purity is Definition 1.9 on page 5.

Since the sentence quantifies over all finite graded posets P, all n, and all k, a single pair (P, k) with $\Delta^{(k-1)}(P)$ contractible and $\Delta^{(k)}(P)$ not contractible settles it, negatively. The witness below has $k = n$; Section 5 says what that does and does not leave open.

2. The witness

Let P be the poset on the ten elements

$$\underbrace{a_0,\ a_1}_{\text{rk}=0} \qquad \underbrace{b_0,\ b_1,\ b_2}_{\text{rk}=1} \qquad \underbrace{c_0,\ c_1,\ c_2}_{\text{rk}=2} \qquad \underbrace{d_0,\ d_1}_{\text{rk}=3}$$

whose cover relations are exactly the following fifteen:

$$\begin{array}{lll} a_0 < b_0 & & \\ a_1 < b_1 & a_1 < b_2 & \\ b_0 < c_0 & b_0 < c_1 & b_0 < c_2 \\ b_1 < c_0 & b_1 < c_1 & \\ b_2 < c_0 & b_2 < c_2 & \\ c_0 < d_0 & & \\ c_1 < d_0 & c_1 < d_1 & \\ c_2 < d_0 & c_2 < d_1 & \end{array} \tag{1}$$

The order on P is the transitive closure of (1). It consists of 31 strict pairs: the fifteen covers, the six pairs $a_i < c_j$ that are present, the six pairs $b_i < d_j$, and all four pairs $a_i < d_j$. Recomputing the cover relation from that closure returns exactly the fifteen pairs of (1), so (1) is a faithful presentation.

Lemma 1. *P is graded of rank 3; it has two minimal elements a_0, a_1 and two maximal elements d_0, d_1, hence neither a minimum nor a maximum, and $\Delta^{(3)}(P) = \Delta(P)$.*

Proof. Every relation in (1) raises the rank by exactly 1; every element of rank < 3 has an upper cover and every element of rank > 0 has a lower cover. Hence every maximal chain of P starts at rank 0, ends at rank 3 and passes through one element of each rank, so it has length 3; there are eleven such chains, and P is graded of rank 3. Only a_0, a_1 have no lower cover and only d_0, d_1 have no upper cover, so these are exactly the minimal and the maximal elements; each of the two sets has two members, so P has neither a minimum nor a maximum. Finally, every chain of P has rank spread at most 3, so $\Delta^{(3)}(P) = \Delta(P)$. ☐

The face numbers of the two filtration stages used below are

$$\Delta^{(2)}(P):\ (10, 27, 18), \qquad \Delta^{(3)}(P):\ (10, 31, 34, 11), \tag{2}$$

with Euler characteristics $+1$ and $+2$.

For $S \subseteq \{0,1,2,3\}$ write $P_S = \{x \in P : \mathrm{rk}(x) \in S\}$ and abbreviate $P_{[i,j]} = P_{\{i,i+1,\dots,j\}}$.

We use one standard fact throughout.

Lemma 2. *Let $X = A \cup B$ be a simplicial complex covered by two subcomplexes with $Y = A \cap B \neq \emptyset$. If A and B are contractible, then X is homotopy equivalent to the suspension ΣY.*

Proof. X is the pushout of the two inclusions $A \hookleftarrow Y \hookrightarrow B$, which are cofibrations of CW pairs, so X is also their homotopy pushout. Replacing A and B by points, which is a homotopy equivalence of pushout diagrams, gives $X \simeq (\{*\} \leftarrow Y \rightarrow \{*\})$-pushout $= \Sigma Y$. □

For a vertex v of a complex X we write $\mathrm{St}_X(v)$ for its closed star, the subcomplex of faces σ with $\sigma \cup \{v\} \in X$, together with their faces. If $X = \Delta(Q)$ is an order complex and $v \in Q$, then $\mathrm{St}_X(v)$ is the cone with apex v over the order complex of $\{x \in Q : x < v \text{ or } x > v\}$, so it is contractible.

3. $\Delta^{(3)}(P)$ is homotopy equivalent to S^2

Proposition 3. $\Delta^{(3)}(P) = \Delta(P) \simeq S^2$. *In particular $\widetilde{H}_2(\Delta(P);\mathbb{Z}) = \mathbb{Z}$ and $\Delta(P)$ is not contractible.*

Proof. The maximal elements of P are d_0 and d_1, and they are incomparable, so no chain contains both, while every chain of P that is maximal ends in d_0 or in d_1. Hence

$$\Delta(P) = \mathrm{St}(d_0) \cup \mathrm{St}(d_1),$$

and both stars are cones, hence contractible. Their intersection is the order complex of $\{x \in P : x < d_0\} \cap \{x \in P : x < d_1\}$. Now $\{x < d_0\} = P_{[0,2]}$, all eight elements of rank ≤ 2, whereas $\{x < d_1\} = P_{[0,2]} \setminus \{c_0\}$: indeed $c_1, c_2 < d_1$ by (1), hence $b_0, b_1 < c_1 < d_1$ and $b_2 < c_2 < d_1$ and $a_0 < b_0 < d_1$, $a_1 < b_1 < d_1$, while $c_0 < d_1$ is not among the relations of P. So

$$Y := \mathrm{St}(d_0) \cap \mathrm{St}(d_1) = \Delta\big(P_{[0,2]} \setminus \{c_0\}\big),$$

the complex on the seven vertices $a_0, a_1, b_0, b_1, b_2, c_1, c_2$ with the eleven edges

$$a_0b_0,\ a_1b_1,\ a_1b_2,\ b_0c_1,\ b_0c_2,\ b_1c_1,\ b_2c_2,\ a_0c_1,\ a_0c_2,\ a_1c_1,\ a_1c_2$$

and the four triangles $a_0b_0c_1$, $a_0b_0c_2$, $a_1b_1c_1$, $a_1b_2c_2$; its Euler characteristic is $7 - 11 + 4 = 0$.

Apply the same argument to Y, at the bottom instead of the top. Every vertex of Y other than a_0, a_1 lies above a_0 or above a_1, so $Y = \mathrm{St}_Y(a_0) \cup \mathrm{St}_Y(a_1)$ with both stars cones. In Y we have $\{x > a_0\} = \{b_0, c_1, c_2\}$ and $\{x > a_1\} = \{b_1, b_2, c_1, c_2\}$, whose intersection is $\{c_1, c_2\}$, two incomparable elements. So $\mathrm{St}_Y(a_0) \cap \mathrm{St}_Y(a_1)$ is a pair of points, i.e. S^0, and Lemma 2 gives $Y \simeq \Sigma S^0 = S^1$. A second application of Lemma 2, now to $\Delta(P) = \mathrm{St}(d_0) \cup \mathrm{St}(d_1)$, gives

$$\Delta(P) \ \simeq\ \Sigma Y \ \simeq\ \Sigma S^1 \ =\ S^2. \qquad \square$$

As a check, $\chi(\Delta(P)) = 10 - 31 + 34 - 11 = 2 = \chi(S^2)$ by (2), whereas a contractible complex has $\chi = 1$.

4. $\Delta^{(2)}(P)$ is collapsible

Recall that a face τ of a simplicial complex X is *free* if it is contained in exactly one proper coface σ; deleting the pair $\{\tau, \sigma\}$ is an *elementary collapse* and does not change the homotopy type. A complex that can be reduced to a single vertex by elementary collapses is *collapsible*, hence contractible.

Proposition 4. *$\Delta^{(2)}(P)$ is collapsible, hence contractible.*

Proof. $\Delta^{(2)}(P)$ has $10 + 27 + 18 = 55$ faces by (2). The following 27 elementary collapses delete 54 of them and leave the single vertex d_1. In each row, τ has exactly one proper coface in the complex remaining at that step, namely σ; the reader may check the rows in order, and each check involves listing the at most three cofaces of a face.

	τ	σ		τ	σ		τ	σ
1	a_0c_0	$a_0b_0c_0$	10	b_0c_0	$b_0c_0d_0$	19	b_2	b_2d_1
2	a_0c_1	$a_0b_0c_1$	11	b_1c_0	$b_1c_0d_0$	20	c_1d_0	$b_0c_1d_0$
3	a_0b_0	$a_0b_0c_2$	12	b_1d_0	$b_1c_1d_0$	21	b_0c_1	$b_0c_1d_1$
4	a_0	a_0c_2	13	b_1c_1	$b_1c_1d_1$	22	c_1	c_1d_1
5	a_1c_1	$a_1b_1c_1$	14	b_1	b_1d_1	23	b_0d_0	$b_0c_2d_0$
6	a_1b_1	$a_1b_1c_0$	15	b_2c_0	$b_2c_0d_0$	24	d_0	c_2d_0
7	a_1c_0	$a_1b_2c_0$	16	c_0	c_0d_0	25	b_0c_2	$b_0c_2d_1$
8	a_1b_2	$a_1b_2c_2$	17	b_2d_0	$b_2c_2d_0$	26	b_0	b_0d_1
9	a_1	a_1c_2	18	b_2c_2	$b_2c_2d_1$	27	c_2	c_2d_1

The face counts run $55, 53, 51, \ldots, 3, 1$, and the last surviving face is the vertex d_1. $\square$

5. The result, and what it does not settle

Theorem 5. *Let P be the ten-element poset of* (1). *Then P is graded of rank* 3*, $\Delta^{(2)}(P)$ is collapsible — in particular contractible — and $\Delta^{(3)}(P)$ is homotopy equivalent to S^2, so it is not contractible. Consequently it is false that $\Delta^{(k)}(P)$ is contractible whenever $\Delta^{(k-1)}(P)$ is.*

Proof. Lemma 1, Proposition 4 and Proposition 3. $\square$

What falls is the universally quantified implication of Remark 2.13, p. 8 of [3], and nothing more. The witness has $k = n = 3$, so the complex that fails to be contractible is the *full* order complex $\Delta^{(n)}(P) = \Delta(P)$; that value of k lies in the range $0 \le k \le n$ of the definition recalled in Section 1 and inside the sentence's quantifier, so the implication is false as stated. It is not a witness at a *proper* filtration stage, and this note leaves untouched the question whether contractibility passes from $\Delta^{(k-1)}(P)$ to $\Delta^{(k)}(P)$ when $k < n$.

No minimality is claimed: neither that ten is the least number of elements of such a witness, nor that $k = 3$ or $n = 3$ is least.

6. Verification

Everything asserted above is determined by the fifteen relations (1), and Sections 3–4 are proofs a reader can follow with no machine. The program `verify.py` accompanying this note, with its recorded output in `verify.output.txt`, reads (1) as printed and re-derives, in exact integer arithmetic and using only the Python standard library, the transitive closure, the cover relation, purity, the f-vectors and Euler characteristics of (2), the integral homology of the filtration stages by Smith normal form, and the intermediate objects of the proof of Proposition 3. The collapse of Section 4 is checked rather than produced: the 27 rows displayed in Proposition 4 are transcribed into the program and replayed, verifying at each row that the printed τ has exactly one proper coface in the complex then remaining and that this coface is the printed σ, and that after row 27 the vertex d_1 alone remains.

The recorded output reports that all 92 of its checks pass. It was produced for a longer draft: its headings follow that draft's numbering, and it also lists checks and closing notes — among them a Mayer–Vietoris computation of the homology of $\Delta^{(2)}(P)$, a discussion of coning this P, a census, and a correction to a remark of [3] — that concern material absent from the present note and on which nothing above depends. In particular nothing in it bears on the open case $k < n$.

References

[1] Y. Di, S. Ivanov, L. Mukoseev, and M. Zhang, *On the path homology of Cayley digraphs and covering digraphs*, J. Algebra **653** (2024), 156–199; arXiv:2305.15683.

[2] R. Kaneta and M. Yoshinaga, *Magnitude homology of metric spaces and order complexes*, Bull. London Math. Soc. **53** (2021), no. 3, 893–905; arXiv:1803.04247.

[3] Y. Kitajima, *Filtered order complexes and magnitude homology of finite graded posets*, preprint, 2026. arXiv:2606.15241. Statement numbers are those printed in version v2 (24 June 2026), whose sectioning begins at Section 0.